# Lab 7: Fuzz Testing for Input Validation Bugs

Software Engineering Lab | Hypothesis property-based testing | Scholarship processor adapted from Lab 6

## Initial run: bugs exposed

The handout names a provided `processor.py`, but no target script accompanied it. This exercise adapts the three scholarship functions from Lab 6 into a standalone target. Initial examples showed impossible marks and attendance being accepted, and negative days remaining being classified as closed.

Before fixes | application output

```
PS> python processor.py
Scholarship application processor
Marks 120 award: 7200
Attendance 150 eligible: True
Negative days-left status: closed
```

Before fixes | Hypothesis failures

```
PS> python -m pytest -q test_processor.py
Hypothesis minimized counterexamples from the initial run:

award_amount_rejects_marks_above_100(marks=101) -> expected ValueError; no error
eligibility_rejects_attendance_above_100(attendance=101) -> expected ValueError; no error
application_status_rejects_negative_days_left(days_left=-1) -> expected ValueError; no error

Additional failures involved floats, None, malformed flags, and invalid strings.
Initial suite: 12 failed, 3 passed.
```

Hypothesis shrank the range bugs to 101 marks, 101 attendance, and -1 day. The initial expanded suite reported 12 failures and 3 passes.

## Fixes and final run

Added checks for integer ranges, known category and income strings, and exact boolean flags. Valid scholarship calculations are retained. The fuzz suite includes valid-input rule properties plus generated boundary and malformed-type inputs for all three functions.

After fixes | application output

```
PS> python processor.py
Scholarship application processor
Marks 120 award: rejected (marks must be between 0 and 100)
Attendance 150 eligibility: rejected (attendance must be between 0 and 100)
Negative days-left status: rejected (days_left must be at least 0)
```

After fixes | Hypothesis tests

```
PS> python -m pytest -q test_processor.py
..................                                       [100%]
18 passed in 1.46s
```

Final result: 18 Hypothesis tests passed. These output captures are formatted from saved command output; full before/after text logs are included in the evidence folder.